Norm-Bounded Federated Updates against Language-Guided Gradient Inversion Attacks

Anonymous ACL submission

Abstract

Vision-language models (VLMs) enable a new class of gradient inversion attacks in federated learning, where a malicious server uses natural-language queries to reconstruct private client data with semantic targeting. We focus on *Geminio* (Shan et al., 2025), the language-guided malicious-model attack of this kind. We make two findings about this threat. First, adversarially robust VLMs designed to defend against input perturbations (Schlarmann et al., 2024; Hossain and Imteaj, 2024; Mao et al., 2023) do not provide privacy protection: they raise Geminio's gradient-amplification loss ratio by 57–61% on medical data, and on UAV imagery four of six robust variants increase mean end-to-end attack F1 above vanilla CLIP (Radford et al., 2021) (3-seed paired t-test reaches $p < 0.05$ for one strengthening direction; the remaining four robust variants fall within seed variance, which we report explicitly). Robustness in this sense is not a substitute for gradient-channel defense. Second, a simple per-update differential-privacy (DP) mechanism we call *Norm-Bounded Federated Updates* (NBFU)—L2 clipping of the entire client update plus calibrated Gaussian noise—substantially reduces attack success and generalizes across three optimization-based reconstruction algorithms (HF-GradInv, IG, and GradInversion). On UAV imagery NBFU at ε=5 drives mean attack F1 to $\leq$ 0.13 across all three attacks (vs. ≤ 0.33 clean); on medical imagery it cuts mean F1 by 50–56% (from 0.67 to 0.29–0.33). Defense on medical leaves a non-trivial residual, and the headline budget (ε=5) destabilizes single-step classifier training, so the defense is best deployed at suspect FL rounds rather than throughout training. A 24-query language sweep shows that the UAV defense is uniform across paraphrase, composition, negation, and language-diversity regimes; medical defense varies by query phrasing. A σ=0 ablation shows that clipping and noise contribute via different mechanisms: clipping disrupts scale-sensitive image-prior regularizers (total variation, BatchNorm statistics), while noise breaks scale-invariant cosine-similarity matching. Removing either component leaves at least one attack family largely undefended. We release configs and code for the four attack methods, the defense, and the reproducible multi-seed sweep.

1 Introduction

Federated learning (FL) promises privacy by replacing data sharing with gradient sharing (McMahan et al., 2017). Gradient inversion attacks (GIAs) undermine that promise by recovering private inputs from shared gradients (Zhu et al., 2019; Geiping et al., 2020; Yin et al., 2021). A recent line of work (Fowl et al., 2022; Wen et al., 2022; Shan et al., 2025) extends GIAs to active malicious servers that craft global model parameters to amplify target gradients before reconstruction.

Shan et al. (2025) introduce a *language-guided* variant in which the server provides a natural-language description of valuable data (e.g., *"any chest X-ray showing pneumonia"*) and a pretrained vision-language model (VLM) (Radford et al., 2021) is used to reshape the malicious model's loss surface so that gradients become dominated by query-matching samples. Crucially, the natural-language query Q is the attacker's only control channel: the same trained classifier and the same client batch yield different reconstructed subsets depending on how Q is phrased. Properties of Q as language—paraphrase invariance, compositional semantics, negation handling, lexical specificity—therefore set the boundary of who is targeted and how cleanly. Because the attacker's discrimination ability scales with VLM quality and with the expressiveness of query language, this class of attack is expected to strengthen as VLM–text alignment improves.

We study this threat from a defense standpoint and report two findings. The first is *negative*: adversarially robust VLMs—a defense category designed to harden CLIP against input perturbations (Mao et al., 2023; Schlarmann et al., 2024; Hossain and Imteaj, 2024)—do not defend against language-guided gradient inversion and in fact strengthen it on medical imagery by 57–61% (Section 5). The second is *positive*: calibrated differentially private FL with per-update gradient clipping (Abadi et al., 2016), which we refer to as Norm-Bounded Federated Updates (NBFU), reduces attack F1 to near zero on UAV imagery and substantially on medical imagery (Section 7). The defense generalizes across the three competitive reconstruction algorithms we test (HF-GradInv (Ye et al., 2024), IG (Geiping et al., 2020), GradInversion (Yin et al., 2021)); a fourth (DLG (Zhu et al., 2019)) is too weak at our image scale to inform defense evaluation (Section 8).

We further give an analytical argument (Section 3) for why clipping is sufficient: it normalizes the gradient norm to a fixed bound C independent of the attacker's loss-ratio amplification, so the post-defense signal-to-noise ratio depends only on C, the calibrated noise scale, and the model dimension—not on how strongly the attacker amplified the target. Measurements of pre-clip gradient norms (15.6 on medical, 527.0 on UAV) match this prediction.

Scope: language as attack surface, not only interface. We position language as an object of study in this attack, not just an interface. The attacker's threat surface depends on language properties of Q: paraphrase variance changes which sample becomes targeted; compositional constraints (*"pneumonia with a portable AP view"*, *"solar panels on residential rooftops"*) sharpen or scatter the targeted subset; negation and exclusion (*"pneumonia without effusion"*) test whether VLM-driven amplification respects truth-functional structure that natural language carries but image embeddings often blur. We treat these as first-class properties of the attack, alongside its image-side and gradient-side properties, and we study a defense that is invariant to all of them by construction.

Contributions.

1. A negative result for VLM-side defense: adversarial robustness in CLIP variants strengthens, rather than mitigates, language-guided gradient inversion.
2. A positive result for gradient-side defense: NBFU at ε=5 (sensitivity-$2C$ calibration; see Section 3) drives mean attack F1 to ≤ 0.13 on UAV across the three reconstruction algorithms we test, and reduces medical F1 by 50–56% on average (3-seed multi-run validation).
3. A mechanistic decomposition via σ=0 ablation: clipping defends attacks with scale-sensitive priors; noise defends scale-invariant cosine matching. Both components are required for attack-agnostic defense.
4. A 24-query language sweep (paraphrase, composition, negation, language-diversity) showing NBFU's defense is uniform across language regime on UAV (mean F1 = 0.06 across 12 queries) but query-dependent on medical (defense margin ranges -113% to $+88\%$ across 12 queries; the F1 threshold metric is noisy at the batch size we use in this domain).
5. A reproducible benchmark covering four GIAs (HF-GradInv, IG, GradInversion, DLG), three batch sizes, three random seeds, and utility curves under NBFU.

2 Background and Threat Model

2.1 Federated learning

We consider FedSGD (federated stochastic gradient descent, McMahan et al., 2017): at each round t the server distributes parameters θ_t; client i samples a private batch B_t^i of size n and shares the averaged gradient

$$G_t^i = \frac{1}{n} \sum_{(x,y)\in B_t^i} \nabla_{\theta_t} \mathcal{L}\big(F_{\theta_t}(x);\, y\big).$$

2.2 Gradient inversion

Optimization-based GIAs (Zhu et al., 2019; Geiping et al., 2020; Yin et al., 2021) reconstruct B_t^i by solving

$$\bar{B}^* = \arg\min_{\bar{B}} \; \delta\big(G_t^i,\; G(\bar{B};\theta_t)\big) + \mathcal{R}(\bar{B}), \quad (1)$$

where δ is a gradient-matching distance (typically Euclidean or cosine) and $\mathcal{R}$ is an image prior (e.g., total variation or feature-space regularization).

2.3 Geminio: language-guided malicious models

Shan et al. (2025) consider an active server that crafts θ_Q such that, for any client batch, samples matching a natural-language query Q have per-sample loss $\mathcal{L}_{\text{target}}$ amplified by a factor R relative to non-matching samples:

$$\mathcal{L}\big(F_{\theta_Q}(x_{\text{target}})\big) \;\approx\; R \cdot \mathcal{L}\big(F_{\theta_Q}(x_{\text{other}})\big).$$

Because per-sample gradient magnitude scales with loss, the averaged client gradient becomes dominated by the target subset, enabling reconstruction via Equation 1 on a effectively single-sample signal even when n is moderately large.

2.4 Threat model

Following Shan et al. (2025); Wen et al. (2022), we consider an active server that may modify θ_t but not the architecture. The server holds a pretrained VLM and an unlabeled auxiliary image set, and observes the client's G_t^i. Notably, the VLM operates on *clean* server-side images, not on adversarial perturbations of client inputs—this distinction motivates our negative result on adversarially robust VLMs (Section 5).

3 Defense: Norm-Bounded Federated Updates

3.1 Definition

Before sharing G, the client applies two operations:

$$G_{\text{clip}} = G \cdot \min\left(1,\ \frac{C}{\|G\|_2}\right), \tag{2}$$

$$G_{\text{def}} = G_{\text{clip}} + \mathcal{N}(0, \sigma^2 I), \tag{3}$$

with $\sigma = 2C \cdot \sqrt{2\ln(1.25/\delta)}/\varepsilon$ following the analytic Gaussian mechanism (Balle and Wang, 2018). We refer to this composite operation as *Norm-Bounded Federated Updates* (NBFU).

Privacy semantics. NBFU performs *per-update* clipping (the entire averaged client update) rather than *per-example* clipping (each per-sample gradient before averaging). We make the DP guarantee precise.

Adjacency. Datasets B, B' are neighboring if one record in the client's batch has been substituted: $|B \triangle B'| = 2$ with $|B| = |B'| = n$.

Mechanism. Let $f(B) = \text{clip}_C\big((1/n)\sum_{(x,y)\in B} \nabla_\theta \mathcal{L}(F_\theta(x); y)\big)$ where $\text{clip}_C(v) = v \cdot \min(1, C/\|v\|_2)$. NBFU releases $\tilde{f}(B) = f(B) + \mathcal{N}(0, \sigma^2 I)$.

Sensitivity. For any neighboring B, B', $\|f(B) - f(B')\|_2 \le 2C$ because $\|f(B)\|_2, \|f(B')\|_2 \le C$ by construction. The bound is tight in the worst case (two clipped vectors on antipodal points of the radius-C ball).

Guarantee. With $\sigma = 2C\sqrt{2\ln(1.25/\delta)}/\varepsilon$, the analytic Gaussian mechanism (Balle and Wang, 2018) gives (ε, δ)-DP for $\tilde{f}$ under the above adjacency, per single FL round. Over T rounds, advanced composition or a moments accountant gives the cumulative budget; for (ε, δ)-DP per round and any slack $\delta' > 0$, advanced composition (Dwork and Roth, 2014, Theorem 3.20) gives $(\varepsilon_T, T\delta + \delta')$-DP with $\varepsilon_T \approx \varepsilon\sqrt{2T\ln(1/\delta')} + T\varepsilon(e^\varepsilon - 1)$ for small ε; a standard choice is $\delta' {=} \delta$, giving total failure probability $\approx (T{+}1)\delta$.

Comparison with record-level DP-SGD. Record-level DP-SGD (Abadi et al., 2016) clips per-example gradients to C before averaging, so the per-record sensitivity scales as C/n and the noise can be calibrated to that. NBFU's per-update bound (sensitivity $2C$) is therefore looser by a factor of $\Theta(n)$ at the same ε. Our motivation for the per-update notion is operational: many FL deployments share only averaged updates, the active malicious-model threat targets recovery within a single round (where per-example clipping is the client's choice, not the server's), and the per-update mechanism requires no changes to existing FL aggregation. We do not claim NBFU is a substitute for record-level DP-SGD when stronger guarantees are needed; it is a deployable defense matched to the Geminio threat model.

Scope: what NBFU is and is not. **Is:** a single-round, per-update aggregate (ε, δ)-DP mechanism against an active malicious server that targets recovery within one FL round. **Adversary capability targeted:** an attacker that crafts θ_Q to amplify query-matching gradients and runs an optimization-based inversion on the released update. **Is not:** record-level DP-SGD ($\Theta(n)$ looser at the same ε), nor client-level DP (the adjacency is record-substitution within one client's batch, not add/remove a client), nor a multi-round budget claim (per-round $\varepsilon{=}5$ composes to a vacuous training-wide budget—see Section 9). The defense effect we measure is the change in attack quality on the released gradient itself; the formal (ε, δ) guarantee is on top of that empirical finding, not in place of it.

3.2 Analytical argument: SNR after defense

Two scopes are relevant. The *optimizer* updates only the 148K-parameter classifier head; the ResNet-18 backbone weights are not modified during training. The *shared update* that the attacker observes, however, is the gradient of the loss with respect to every parameter that has `requires_grad=True`—the full $d \approx 1.13\times10^7$ parameter set, since we leave the backbone unfrozen even though it is not optimized (76% of the gradient ℓ_2 norm is in the backbone, see Section 7.2). Inversion attacks search the full d-dimensional space, so the relevant d for both the noise norm and the signal-to-noise ratio (SNR) bound is the larger one. The expected ℓ_2 norm of the noise term in Equation 3 is $\|\mathcal{N}\| \approx \sigma\sqrt{d}$. After clipping, the signal norm satisfies $\|G_{\text{clip}}\|_2 \leq C$ regardless of $\|G\|$. Hence the post-defense SNR satisfies

$$\text{SNR} \leq \frac{C}{\sigma\sqrt{d}}. \tag{4}$$

This bound is independent of the attacker's amplification factor R. As Geminio's loss ratio grows, the unclipped $\|G\|$ grows, but clipping renormalizes the signal to scale C and the calibrated noise is set to C as well, so R drops out.

Two complementary defense components. Equation 4 captures the noise side of the argument: small SNR is sufficient. Clipping plays a separate role, which Equation 4 alone does not show: it bounds sensitivity (enabling the DP analysis above) and it disrupts reconstruction objectives whose value depends on *absolute* gradient magnitude—specifically Euclidean matching and image-prior regularizers (total variation, BatchNorm-statistic) that are scale-sensitive. Pure cosine-similarity matching is scale-invariant, so against such attacks the noise does the entire defense work. Section 7.2 dissects which component is doing what, attack by attack, via a σ=0 ablation.

Numerical SNR. For our setup (C=1, ε=5, δ=10^{-5}), $\sigma \approx 1.94$ (using sensitivity $2C$) and Equation 4 gives SNR $\approx 1.5\times10^{-4}$: the defended gradient is dominated by noise. Empirical gradient norms (Table 1) confirm the regime.

4 Experimental Setup

Datasets and tasks. We test on two image classification benchmarks: **ChestMNIST** (Yang et al., 2023) (15-way, 28×28 chest X-rays upsampled to 224×224) and **UAVScenes** (Wang et al., 2025) (18-way, 224×224 aerial drone imagery).

Model. A ResNet-18 (He et al., 2016) backbone (ImageNet-pretrained) with a 3-layer classifier head (512→256→64→ N classes). The optimizer updates only the classifier head (148K parameters), but the backbone keeps `requires_grad=True` so gradients propagate through it; the *shared* update consequently spans the full $d \approx 1.13\times10^7$ parameters and the attacker uses them all (76% of $\|G\|^2$ resides in the backbone). Our main experiments use `model.eval()` for the victim forward/backward (BatchNorm uses fixed running statistics), following the convention in Geiping et al. (2023). Recent systematizations note that this configuration can simplify inversion relative to a realistic training-mode client. We therefore include a BN `train`-mode robustness check (Section 8.2, Appendix A) and find that, while train mode lowers the clean attack on medical (0.500 vs. 0.667 in eval), the NBFU defense margin is actually larger in train mode.

VLMs. Vanilla CLIP ViT-L/14 (Radford et al., 2021), BiomedCLIP (Zhang et al., 2023), and adversarially robust variants SimCLIP-2/4 (Hossain and Imteaj, 2024), FARE-2/4 (Schlarmann et al., 2024), and TeCoA-2/4 (Mao et al., 2023). The numeric suffix denotes the adversarial training budget ($\epsilon = k/255$).

Reconstruction algorithms. We use four optimization-based GIAs from Geiping et al. (2023): HF-GradInv (Ye et al., 2024) (Geminio's default), IG (Geiping et al., 2020), GradInversion (Yin et al., 2021, 2020), and DLG (Zhu et al., 2019; Zhao et al., 2020). All run for 24,000 iterations with seed-controlled random initialization.

Defense. NBFU as defined in Equations 2–3, with clipping norm C=1.0, δ=10^{-5}, batch size n=8 unless stated otherwise. We sweep $\varepsilon \in \{0.1, 1, 5, 10, 50\}$.

Metrics. **Attack F1** at cosine threshold 0.90 between reconstruction and target output-layer gradients (the metric used in Shan et al. (2025)); **LPIPS** (Zhang et al., 2018), lower is better; and PSNR. We report mean ± standard deviation over three seeds (42, 123, 256). All comparisons in tables are at the same seed unless aggregated.

5 Robust VLMs Do Not Help Defense

We train Geminio's malicious model for 5 medical and 6 UAV queries with each of seven VLM encoders, evaluating via two metrics: *loss ratio*

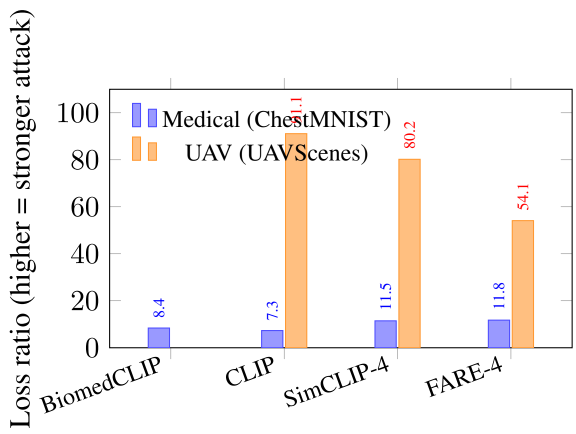

Figure 1: Geminio loss ratio across VLM variants. On medical, robust VLMs SimCLIP-4 and FARE-4 *increase* amplification by 57–61% over vanilla CLIP. On UAV, vanilla CLIP already attains 91×; robust variants do not improve.

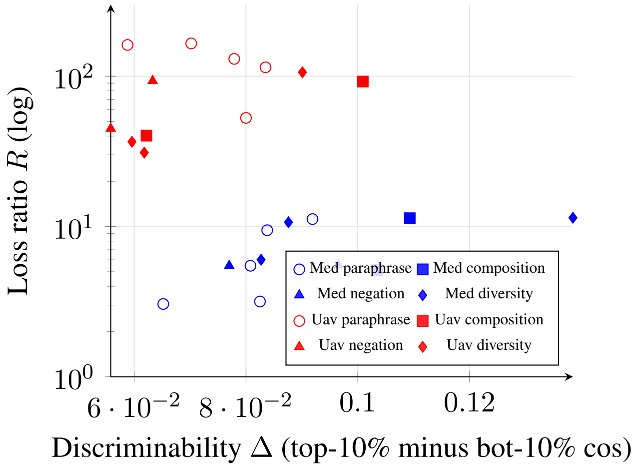

Figure 2: Discriminability $\Delta(Q)$ vs. loss ratio $R(Q)$ across 24 queries (blue=medical/BiomedCLIP, red=UAV/CLIP; markers by class). The relationship is non-monotone within each domain: a VLM-only metric is not a reliable forecast of post-training amplification.

(top-10%-vs-bottom-90% loss after Phase 2 training, the metric used by Shan et al. (2025)) and end-to-end *attack F1* (HF-GradInv reconstruction, mean over 3 seeds). The two often disagree (full table in Appendix G).

On medical, robust VLMs raise loss-ratio by 57–61% over vanilla CLIP but *weaken* the end-to-end attack F1 (every robust variant below CLIP's 0.750; paired t-test against CLIP, df=2: significant for 3 of 6 at $p < 0.05$, marginal for 2 more). LPIPS confirms (CLIP 0.93 vs. robust 0.55–0.69). On UAV the F1 picture is noisier: only 2 of 6 reach paired-t significance (SimCLIP-2 weakens, TeCoA-4 strengthens); the remaining four fall within seed variance. Geminio's loss ratio measures how sharply the model concentrates gradient energy on a target; F1 also depends on whether that targeted pattern is invertible—robust encoders sharpen targeting but on medical also yield reconstructions perceptually closer to the truth, failing to defend on both axes simultaneously. The implication is a threat-model mismatch: SimCLIP/FARE/TeCoA defend against *adversarial inputs*, but Geminio uses the VLM on *clean* server-side images. VLM-side robustness is not a substitute for gradient-channel defense.

6 Language Properties of the Attack Surface

Geminio's threat surface is governed by language: the same malicious-server infrastructure, auxiliary set, and victim batch yield different reconstructable subsets depending on how the query Q is phrased. We sweep 24 queries (12 medical, 12 UAV) along four axes—**paraphrase** (5 variants per domain), **composition** (and/or), **negation** (exclusion and pure), and **language diversity** (Spanish, typo-injected, extra-verbose)—and measure three quantities per query: loss ratio $R(Q)$, discriminability margin $\Delta(Q) = \mathbb{E}_{\text{top-10\%}} \cos(e_i, t_Q) - \mathbb{E}_{\text{bot-10\%}} \cos(e_i, t_Q)$ on the auxiliary set, and 3-seed mean HF-GradInv attack F1 (clean and NBFU ε=5). Full per-query table is in Appendix D.

Three findings. (1) *Loss ratio and F1 disagree.* On medical, R varies $\approx$ 4× (3.0–11.4) but F1 varies $\approx$ 7× (0.13–0.92); `pna_synonym` and `pna_indirect` have nearly identical R (3.2 vs. 3.1) but F1 differs 3.5× (0.13 vs. 0.46). Targeting sharpness and gradient invertibility are independent quantities. (2) *Domain VLM vs. general VLM diverge under perturbation.* BiomedCLIP is robust to typos (R=11.4) and Spanish (R=6.0); vanilla CLIP collapses on both (typos: 31, Spanish: 37). Composition (`compAND`) sharpens medical but weakens UAV; disjunction and pure negation weaken targeting on both, yet `sol_negPURE` ("does not show solar panels") still yields the *highest* UAV F1 (0.50)—the strongest evidence that R and F1 measure different things. (3) *NBFU holds uniformly on UAV, varies by query on medical.* UAV NBFU F1 $\leq$ 0.125 across *all* 12 queries (mean 0.06), invariant to the language regime. Medical defense margins range from

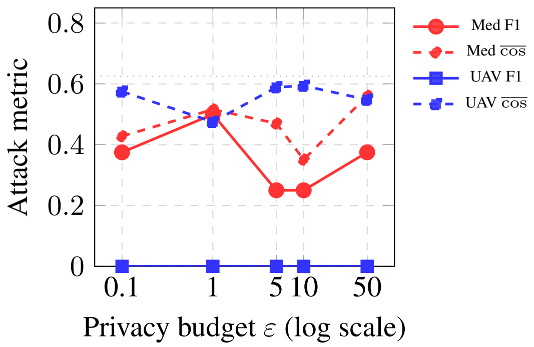

Figure 3: Attack F1 (solid) and average sample-level gradient cosine similarity $\overline{\cos}$ (dashed) vs. NBFU privacy budget ε (HF-GradInv, batch size 8, seed 42, sensitivity $2C$). UAV F1 is 0 across the range; medical F1 is non-monotonic but $\overline{\cos}$ stays well below the no-defense value of 0.92.

$+88\%$ (pna_negPURE: 0.333→0.042) to -113% (pna_compAND: 0.333→0.708); 6 of 12 queries see NBFU < clean. The F1 threshold at $n{=}8$ is noisy (± 0.10–0.18 across seeds for many cells), and visually-similar medical reconstructions can satisfy the cos-0.90 threshold through noise interactions. Defense robustness on visually-similar domains should be evaluated by language stress-testing, not a single canonical query.

7 Norm-Bounded Federated Updates Reduce the Attack

7.1 Privacy-budget sweep

Figure 3 shows attack F1 and average per-sample gradient cosine similarity as a function of ε for HF-GradInv on the canonical Geminio queries (sensitivity-$2C$ calibration (Section 3)). NBFU defends across the full range $\varepsilon \in [0.1, 50]$. On UAV, F1 stays at 0 for every ε tested. On medical, F1 sits in 0.25–0.50 across the range; while it fluctuates non-monotonically (the threshold metric is sensitive to single-sample crossings at $n{=}8$), the continuous $\overline{\cos}$ stays in 0.35–0.56 (vs. 0.92 at clean), confirming that the underlying gradient signal is consistently degraded.

7.2 Mechanism: which component is doing what

Empirical pre-clip gradient norms set the scale at which clipping operates:

Domain	$\|G\|$ before clip	$C/\|G\|$
Medical	15.6	0.064
UAV	527.0	0.0019

Table 1: Pre-clip gradient norms (HF-GradInv, $C{=}1$). Geminio's amplification is much larger on UAV (loss ratio $\sim$91×) than on medical ($\sim$8×); clipping removes most of that amplification.

Decomposing the defense via $\sigma{=}0$. To separate clipping from noise, we run a clipping-only condition (apply Equation 2 but skip Equation 3). Full per-attack results are in Appendix B, Table 6; the headline pattern is two regimes. On UAV with IG (cosine matching with TV), clipping alone has zero effect (0.208 clean = 0.208 clip-only)—cosine matching is scale-invariant—and defense comes entirely from noise (clip+noise → 0.042). For medical IG/GradInversion (scale-sensitive image priors), clipping alone provides 13–25% of the F1 reduction by changing the relative weighting between gradient-matching loss and prior; noise on top closes the remaining gap. The common intuition that "clipping bounds sensitivity, noise provides DP" is correct but incomplete: in the GIA setting both components contribute empirically via different mechanisms, and removing either leaves at least one attack family largely undefended.

7.3 Comparison with other defenses

Table 7 (Appendix C) compares NBFU with the standard defenses tested in Shan et al. (2025) and with VLM-side robust encoders. NBFU $\varepsilon{=}5$ is the strongest reliable defense (Med F1 0.292, UAV 0.125, multi-seed); NBFU $\varepsilon{=}10$ is the strongest *trainable+defended* regime (Med 0.333, UAV 0.042) with single-step training stability preserved. *Combining defenses is non-monotone*: FedAvg-5+NBFU $\varepsilon{=}5$ achieves full UAV defense (0.000) but raises Medical to 0.500, worse than either alone—the FedAvg local-update's larger pre-clip norm interacts with fixed $C{=}1$ in a way that the NBFU calibration was not designed for. Pruning/Laplace defenses are inconsistent (also reported in Shan et al. (2025)); robust VLMs are unreliable per Section 5.

8 Defense Generalizes Across Reconstruction Algorithms

We rerun the $\varepsilon{=}5$ defense against three additional reconstruction algorithms: IG (Geiping et al., 2020) (cosine matching with total-variation prior), GradInversion (Yin et al., 2021) (Euclidean matching with TV, norm, and BatchNorm-statistic priors),

Domain	Attack	Clean F1	NBFU ε=5 F1
Medical	HF-GradInv	0.667±.07	0.292±.07
	IG	0.667±.07	0.292±.07
	GradInversion	0.667±.07	0.333±.14
UAV	HF-GradInv	0.250±.13	0.125±.22
	IG	0.208±.07	0.042±.07
	GradInversion	0.333±.14	0.000±.00

Table 2: Multi-seed defense generalization, sensitivity-$2C$ calibration, 3 seeds (42, 123, 256), 24k iterations. The remaining medical residual reflects (i) moderate loss-ratio amplification ($\sim$8$\times$) leaving less for clipping to remove, (ii) image-prior regularizers (TV, BN-statistic) providing candidate-side signal independent of the gradient channel, and (iii) F1 variance at n=8.

and DLG (Zhu et al., 2019) (Euclidean matching, no prior). We also include the original HF-GradInv (Ye et al., 2024) for comparison. Three seeds (42, 123, 256), 24,000 iterations.

8.1 Multi-seed results

Table 2 reports mean ± std over 3 seeds for all six (domain, attack) cells. NBFU ε=5 reduces F1 in every cell, with the defense effect substantially exceeding seed-to-seed variance. Medical reductions are 50–56% across attacks; UAV defense ranges 50–100% (full defense for GradInversion).

8.2 Robustness checks

Three sensitivity sweeps (full tables in Appendix A) confirm that the qualitative conclusions hold across plausible configurations. (i) **Clipping bound C:** sweeping $C \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$ at ε=5 produces a U-shaped medical F1 curve (small or large C defend; intermediate $C\approx 2$ is the weakest regime, F1=0.50). UAV defense is near-saturated across the range. (ii) **BatchNorm train mode:** switching the victim from `eval` to `train` mode (BN uses batch statistics, per the systematization recommendation) *strengthens* the medical defense margin from 56% to 92%; UAV is unchanged. (iii) **Batch size:** at $n \in \{8, 16, 32\}$ the relative medical defense margin shrinks from 50% to 7% because the unclipped gradient norm decreases with n; UAV remains fully defended.

(iv) **Frozen backbone:** freezing the ResNet-18 backbone so only the 148K head-gradient is shared preserves or improves NBFU's defense margin (medical 58% reduction, UAV 83% at ε=5; full details in Appendix E). Percentile-based adaptive clipping (Andrew et al., 2021) would close the U-shape from (i) by avoiding the intermediate-C trough; we leave the adaptive procedure to future work.

9 Privacy–Utility Trade-off

A defense is only deployable if the underlying training task remains learnable. We train the ResNet-18 classifier head on ChestMNIST under NBFU at each ε for 10 epochs at batch size 64, applying NBFU to the gradient at every optimizer step (single-client FL simulation). ε is *per-step*; cross-step composition via the moments accountant yields a vacuous training-wide budget at per-step ε=5, so this is a stability stress test rather than a tight end-to-end privacy claim. Figure 5 and Table 3 show a sharp cliff at $\varepsilon\approx$5–10: at $\varepsilon \geq 10$ test accuracy matches the no-defense baseline (52.5%); at $\varepsilon \leq 5$ training is unstable (loss diverges and accuracy oscillates). Pure clipping (no noise) does not hurt utility (52.8%): the destabilizing component is the noise magnitude, not the clipping bound.

Defense	Test acc	Trainable
None (FedSGD)	0.525	yes
Clip-only (σ=0)	0.528	yes
NBFU ε=50	0.531	yes
NBFU ε=10	0.530	yes
NBFU ε=5	0.121	**no**
NBFU ε=1	0.339	**no**
NBFU ε=0.1	0.284	**no**

Table 3: Test accuracy on ChestMNIST after 10 epochs of classifier-head training under NBFU at each ε. Training is unstable at $\varepsilon \leq 5$: loss diverges and accuracy oscillates between near-zero and majority-class baseline. See Figure 5 in appendix.

Reconciling defense and utility. The strongest defense regime ε=5 (Section 7) is incompatible with stable training, but the trainable+defended regime ε=10 still defends substantially: 3-seed mean F1 = 0.333 (medical, 50% reduction from clean 0.667) and 0.042 (UAV, 83% reduction from 0.250), at test acc 53.0% vs. 52.5% baseline. A practitioner can deploy NBFU ε=10 always-on, switch to ε=5 at suspect rounds (Geminio attacks single rounds), or apply NBFU only after convergence. Adaptive C schemes (Andrew et al., 2021) would move the cliff but were not optimized in this study. Note that the utility cliff reflects noise added across the 148K-parameter classifier subspace; NBFU applied where the optimizer also updates the 11.3M backbone parameters would face proportionally larger utility costs.

The "suspect-rounds" lever needs a client-side detector; candidate signals include parameter-divergence between consecutive received models, per-sample loss skew on a client holdout, and output-layer weight anomalies (Appendix F). A full study is left to future work.

10 Related Work

Gradient inversion attacks. DLG (Zhu et al., 2019) introduced gradient leakage as a privacy threat; iDLG (Zhao et al., 2020) added analytic label recovery. Geiping et al. (2020) replaced Euclidean distance with cosine similarity and added total-variation regularization, scaling the attack to ImageNet. GradInversion (Yin et al., 2021) added BatchNorm-statistic and feature-space priors for batch reconstruction. These remain the workhorse algorithms used as the reconstruction step in active-server attacks.

Active-server attacks. Robbing the Fed (Fowl et al., 2022) introduced architecture-modification attacks; Fishing (Wen et al., 2022) amplified target gradients via output-layer parameter scaling. Geminio (Shan et al., 2025) generalized targeting to natural-language queries via VLMs. Our work focuses on defenses against this last family.

DP defenses. Differential privacy in deep learning was formalized by Abadi et al. (2016); Dwork and Roth (2014) provide a foundation. McMahan et al. (2018) introduced *client-level* DP for FL with adjacency at the level of whole clients; our per-update-aggregate notion is closer to the per-round protections analyzed by Zhang et al. (2021), who study convergence under per-update clipping in heterogeneous FL and provide guidance on threshold selection (we discuss their results in Section 7.2). Andrew et al. (2021) introduced percentile-based adaptive clipping for differentially private FL; Zhang et al. (2024) combine differentially private sketches with adaptive clipping for communication efficiency; and de Retana et al. (2025) empirically corroborate that calibrated DP-SGD reduces gradient inversion success on large pretrained models—consistent with our findings, though their evaluation does not include language-guided active attacks.

Non-DP defenses. Shan et al. (2025) themselves test gradient pruning, additive noise, and FedAvg, and report these are inconsistent or insufficient; our results confirm this for pruning and noise. Beyond DP, recent non-DP defenses include SVDefense (Luo et al., 2025), which projects gradients onto a low-rank subspace to remove inversion-relevant directions, and the convolutional variational bottleneck of Scheliga et al. (2023), which adds client-side variational noise driven by the model rather than a privacy budget. These approaches are complementary: NBFU provides a formal DP guarantee at known utility cost, while SVDefense and variational-bottleneck defenses target the inversion attacker without claiming DP. A comparison study at matched empirical defense level is left to future work.

Robust VLMs. TeCoA (Mao et al., 2023), FARE (Schlarmann et al., 2024), and Sim-CLIP (Hossain and Imteaj, 2024) adversarially fine-tune CLIP's vision encoder against input perturbations. Our results (Section 5) show that the threat model mismatch with Geminio means robustness in this sense does not transfer to FL gradient privacy: it raises loss-ratio amplification on medical and raises end-to-end attack F1 on UAV.

11 Conclusion

We studied defenses against Geminio, a language-guided gradient inversion attack that uses VLMs to target valuable client data with natural-language queries. We showed that adversarially robust VLMs do not protect against this attack: they raise Geminio's gradient-amplification loss ratio by 57–61% on medical data, and on UAV imagery four of six robust variants raise mean end-to-end attack F1 above vanilla CLIP (multi-seed). NBFU—per-update L2 clipping plus calibrated Gaussian noise, providing (ε, δ)-DP at the per-update aggregate—substantially reduces the attack across three reconstruction algorithms, driving mean F1 to zero on UAV at ε=5 and cutting medical F1 by half. A σ=0 ablation reveals that clipping and noise defend through complementary mechanisms: clipping disrupts scale-sensitive image priors, noise breaks scale-invariant cosine matching. The strongest defense regime (ε=5) destabilizes single-step training in our setup; we discuss round-targeted deployment as a path to reconcile defense and utility. As VLMs continue to scale, gradient-level DP appears more durable than encoder-level robustness as a privacy mechanism in federated learning, provided both clipping and noise are deployed together.

Limitations

We highlight eight limitations.

First, we evaluate on a single architecture (ResNet-18). The SNR argument depends on d, the parameter count; very small models with $d \lesssim 10^5$ would have substantially higher SNR after defense and could be more reconstructable.

Second, our reconstruction quality metric, attack F1 at cosine threshold 0.90, is coarse at small batch sizes (n=8 admits only 9 discrete values). We report the continuous *average* cosine similarity alongside F1 in Section 7.2 to address metric variance, but a richer evaluation would include per-sample reconstruction quality histograms.

Third, we test three GIAs (DLG, IG, GradInversion) at full-layer scale and one (HF-GradInv) at all batch sizes; the former three exceed GPU memory at $n \geq 16$ in our setup. The batch-size sweep is therefore reported only for HF-GradInv.

Fourth, the SNR bound (Equation 4) is an informal expectation argument, not a high-probability bound on the reconstruction error. A rigorous treatment would account for the directional structure of the reconstruction objective and the noise interaction with image priors.

Fifth, we evaluate the privacy–utility curve under NBFU on classifier-head training only (Section 9); a full deployment that trains the backbone under NBFU may face larger utility costs due to the higher-dimensional clipping target.

Sixth, our DP guarantee is per-update aggregate (ε, δ) in the substitute-one-record neighboring relation, with sensitivity $2C$. Tighter record-level guarantees would require per-example clipping; presence/absence client-level DP would require sampling clients. We do not report cross-round composition; deployments must compose the per-round budget over training using a standard accountant.

Seventh, our attack coverage is restricted to gradient-matching GIAs (DLG, IG, GradInversion, HF-GradInv). Four recent attacks lean on stronger priors: GIFD (Fang et al., 2023) uses a pretrained GAN; GUIDE (Carletti et al., 2025) uses a learned denoiser to clean reconstructed images; GI-SMN (Qian et al., 2024) uses a style-migration network to escape the prior-knowledge requirement; and GI-NAS (Yu et al., 2025) adaptively searches network architectures to exploit implicit priors. Public reference code for the latter two was not available at submission time, and the former two require domain-specific pretrained generative models (no such models exist off-the-shelf for ChestMNIST X-rays or UAVScenes aerial imagery). Whether NBFU's defense margin holds against generative-prior or NAS-driven attacks is left to future work. The σ=0 mechanism analysis predicts that the noise component should still degrade these attacks, since they rely on a gradient-matching residual to drive image-prior optimization, and the SNR bound in Equation 4 applies to any attacker reading the released gradient—regardless of the prior they pair it with.

Eighth, we evaluate the attack and defense under FedSGD with per-client, per-round gradient transmission visible to the server. Two operational variants that change attack feasibility are not tested here. *Secure aggregation* (Bonawitz et al., 2017) hides individual client updates from the server, summing them via cryptographic protocols; Geminio in its original form requires per-client tailoring of θ_Q (the same model goes to all clients via the protocol's broadcast), so an active server using secure aggregation must rely on aggregate-level reconstruction. NBFU's per-update DP guarantee at the client level remains meaningful in this setting (the aggregated update is still a sum of NBFU-protected client updates), but the attacker's reconstruction-from-sum problem is harder than the per-client case we evaluate. *FedAvg with multiple local epochs* extends the client-side computation between rounds; we report FedAvg defense results at 5 and 20 local epochs and a FedAvg-5+NBFU combination in Table 7. The combination is non-monotone: on UAV it is the strongest defense in the table (F1=0.000), but on medical it raises F1 to 0.500—worse than either component alone. A clean defense extension for multi-epoch FedAvg would set C adaptively as a percentile of the local-update norm distribution (rather than the fixed C=1 we use for single-step gradients) and apply the analytic Gaussian mechanism to the full local-update vector; characterizing the precise calibration mismatch and adaptive recipe is left to future work. Implementation note: we caught and fixed a state-restoration bug in our FedAvg simulator (BN running buffers were not restored after the local training pass, which artificially inflated the F1 metric); reported numbers reflect the fix.

Ethical Considerations

This paper studies attacks and defenses in federated learning. Our defense, NBFU, is a constructive contribution: it provides a deployable mitigation for an existing published attack (Shan et al., 2025). We make available no new attack capabilities beyond reproducing baselines from the literature for evaluation purposes. Our evaluation uses public benchmarks (ChestMNIST, UAVScenes); no private user data is collected or distributed.

A subtle ethical concern is our negative result on robust VLMs. We disclose this finding to prevent misallocation of defense engineering effort: practitioners should not assume that adversarially robust encoders, designed for a different threat model, transfer protection to federated gradient privacy. The mechanism we identify (Section 5) suggests this is a structural property, not an artifact of specific encoder weights.

References

Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. 2016. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 308–318.

Galen Andrew, Om Thakkar, Brendan McMahan, and Swaroop Ramaswamy. 2021. Differentially private learning with adaptive clipping. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:1905.03871.

Borja Balle and Yu-Xiang Wang. 2018. Improving the Gaussian mechanism for differential privacy: Analytical calibration and optimal denoising. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 394–403. ArXiv:1805.06530.

Keith Bonawitz, Vladimir Ivanov, Ben Kreuter, Antonio Marcedone, H. Brendan McMahan, Sarvar Patel, Daniel Ramage, Aaron Segal, and Karn Seth. 2017. Practical secure aggregation for privacy-preserving machine learning. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 1175–1191.

Vincenzo Carletti, Pasquale Foggia, Carlo Mazzocca, Giuseppe Parrella, and Mario Vento. 2025. GUIDE: Enhancing gradient inversion attacks in federated learning with denoising models. *arXiv preprint arXiv:2510.17621*.

Miguel Fernandez de Retana, Unai Zulaika, Rubén Sánchez-Corcuera, and Aitor Almeida. 2025. Differential privacy: Gradient leakage attacks in federated learning environments. *arXiv preprint arXiv:2510.23931*.

Cynthia Dwork and Aaron Roth. 2014. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3-4):211–407.

Hao Fang, Bin Chen, Xuan Wang, Zhi Wang, and Shu-Tao Xia. 2023. GIFD: A generative gradient inversion method with feature domain optimization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.

Liam Fowl, Jonas Geiping, Wojciech Czaja, Micah Goldblum, and Tom Goldstein. 2022. Robbing the fed: Directly obtaining private data in federated learning with modified models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2110.13057.

Jonas Geiping, Hartmut Bauermeister, Hannah Dröge, and Michael Moeller. 2020. Inverting gradients — how easy is it to break privacy in federated learning? In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2003.14053.

Jonas Geiping, Liam Fowl, and Yuxin Wen. 2023. Breaching: A framework for attacks against privacy in federated learning. `https://github.com/JonasGeiping/breaching`. Software framework collecting gradient-inversion and active-server attacks.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778.

Md Zarif Hossain and Ahmed Imteaj. 2024. Sim-CLIP: Unsupervised siamese adversarial fine-tuning for robust and semantically-rich vision-language models. *arXiv preprint arXiv:2407.14971*.

Chenxiang Luo, David K. Y. Yau, and Qun Song. 2025. SVDefense: Effective defense against gradient inversion attacks via singular value decomposition. *arXiv preprint arXiv:2510.03319*.

Chengzhi Mao, Scott Geng, Junfeng Yang, Xin Wang, and Carl Vondrick. 2023. Understanding zero-shot adversarial robustness for large-scale models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2212.07016.

Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. 2017. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 1273–1282.

H. Brendan McMahan, Daniel Ramage, Kunal Talwar, and Li Zhang. 2018. Learning differentially private recurrent language models. In *International Conference on Learning Representations (ICLR)*. ArXiv:1710.06963.

Jin Qian, Kaimin Wei, Yongdong Wu, Jilian Zhang, Jipeng Chen, and Huan Bao. 2024. GI-SMN: Gradient inversion attack against federated learning without prior knowledge. *arXiv preprint arXiv:2405.03516*.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 8748–8763.

Daniel Scheliga, Patrick Mäder, and Marco Seeland. 2023. Privacy preserving federated learning with convolutional variational bottlenecks. *arXiv preprint arXiv:2309.04515*.

Christian Schlarmann, Naman Deep Singh, Francesco Croce, and Matthias Hein. 2024. Robust CLIP: Unsupervised adversarial fine-tuning of vision embeddings for robust large vision-language models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*. ArXiv:2402.12336.

Junjie Shan, Ziqi Zhao, Jialin Lu, Rui Zhang, Siu Ming Yiu, and Ka-Ho Chow. 2025. Geminio: Language-guided gradient inversion attacks in federated learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. ArXiv:2411.14937.

Sijie Wang, Siqi Li, Yawei Zhang, Shangshu Yu, Shenghai Yuan, Rui She, Quanjiang Guo, JinXuan Zheng, Ong Kang Howe, Leonrich Chandra, Shrivarshann Srijeyan, Aditya Sivadas, Toshan Aggarwal, Heyuan Liu, Hongming Zhang, Chujie Chen, Junyu Jiang, Lihua Xie, and Wee Peng Tay. 2025. UAVScenes: A multi-modal dataset for UAVs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. ArXiv:2507.22412.

Yuxin Wen, Jonas Geiping, Liam Fowl, Micah Goldblum, and Tom Goldstein. 2022. Fishing for user data in large-batch federated learning via gradient magnification. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*. ArXiv:2202.00580.

Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. 2023. MedMNIST v2 — a large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41.

Zipeng Ye, Wenjian Luo, Qi Zhou, and Yubo Tang. 2024. High-fidelity gradient inversion in distributed learning. In *Proceedings of the 38th AAAI Conference on Artificial Intelligence*, pages 19983–19991. Referred to as HF-GradInv in this paper; implementation at github.com/MiLab-HITSZ/2023YeHFGradInv.

Hongxu Yin, Arun Mallya, Arash Vahdat, Jose M. Alvarez, Jan Kautz, and Pavlo Molchanov. 2021. See through gradients: Image batch recovery via GradInversion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16337–16346.

Hongxu Yin, Pavlo Molchanov, Jose M. Alvarez, Zhizhong Li, Arun Mallya, Derek Hoiem, Niraj K. Jha, and Jan Kautz. 2020. Dreaming to distill: Data-free knowledge transfer via DeepInversion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8715–8724.

Wenbo Yu, Hao Fang, Bin Chen, Xiaohang Sui, Chuan Chen, Hao Wu, Shu-Tao Xia, and Ke Xu. 2025. GI-NAS: Boosting gradient inversion attacks through adaptive neural architecture search. *IEEE Transactions on Information Forensics and Security (TIFS)*. ArXiv:2405.20725.

Meifan Zhang, Zhanhong Xie, and Lihua Yin. 2024. Private and communication-efficient federated learning based on differentially private sketches. *arXiv preprint arXiv:2410.05733*.

Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. 2018. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 586–595.

Sheng Zhang, Yanbo Xu, Naoto Usuyama, Jaspreet Bagga, Robert Tinn, Sam Preston, Rajesh Rao, Mu Wei, Naveen Valluri, Cliff Wong, Matthew P. Lungren, Tristan Naumann, and Hoifung Poon. 2023. BiomedCLIP: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. *arXiv preprint arXiv:2303.00915*.

Xinwei Zhang, Xiangyi Chen, Mingyi Hong, Zhiwei Steven Wu, and Jinfeng Yi. 2021. Understanding clipping for federated learning: Convergence and client-level differential privacy. *arXiv preprint arXiv:2106.13673*.

Bo Zhao, Konda Reddy Mopuri, and Hakan Bilen. 2020. iDLG: Improved deep leakage from gradients. *arXiv preprint arXiv:2001.02610*.

Ligeng Zhu, Zhijian Liu, and Song Han. 2019. Deep leakage from gradients. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 14747–14756.

C	Med F1	Med $\overline{\cos}$	UAV F1	UAV $\overline{\cos}$
0.1	0.250	0.355	0.125	0.659
0.5	0.375	0.475	0.125	0.711
1.0	0.250	0.343	0.000	0.625
2.0	0.500	0.574	0.125	0.680
5.0	0.125	0.338	0.000	0.563

Table 4: Attack F1 vs. clipping bound C at fixed ε=5 (HF-GradInv, seed 42, sensitivity-$2C$ noise calibration). The medical F1 row is U-shaped because σ scales with C: small C defends via tight clipping; large C via large noise; an intermediate $C\approx2$ leaves F1 at 0.50.

Domain	BN mode	Clean	NBFU ε=5	Drop
Med.	eval	0.667±.07	0.292±.07	56%
	train	0.500±.00	0.042±.06	92%
UAV	eval	0.250±.13	0.125±.22	50%
	train	0.250±.00	0.125±.10	50%

Table 5: HF-GradInv at ε=5 (sensitivity $2C$), mean±std over 3 seeds (42, 123, 256). Train mode lowers the clean attack on medical (0.667→0.500) and produces a *larger* defense margin (92% vs. 56% drop). UAV is invariant to BN mode.

A Robustness check details

A.1 Clipping bound C sensitivity (Section 8.2)

A.2 BatchNorm train-mode multi-seed (Section 8.2)

A.3 Batch-size sweep (Section 8.2)

A.4 Privacy–utility curve (Section 9)

B σ=0 mechanism ablation

Full per-attack mechanism decomposition for Section 7.2.

Domain	Attack	Clean	Clip-only	NBFU ε=5
Med.	HF-GradInv	0.667±.07	0.625±.00	0.292±.07
	IG	0.667±.07	0.583±.19	0.292±.07
	GradInv	0.667±.07	0.500±.00	0.333±.14
UAV	HF-GradInv	0.250±.13	0.125±.13	0.125±.22
	IG	0.208±.07	0.208±.07	0.042±.07
	GradInv	0.333±.14	0.125±.13	0.000±.00

Table 6: Clipping-only (σ=0) vs. full NBFU (ε=5), mean±std over 3 seeds. Clipping handles attacks with scale-sensitive image priors (IG/GradInversion on medical drop 13–25% under clipping alone); noise handles scale-invariant cosine matching (UAV IG sees zero effect from clipping but full defense once noise is added).

C Defense comparison

Full comparison for Section 7.3.

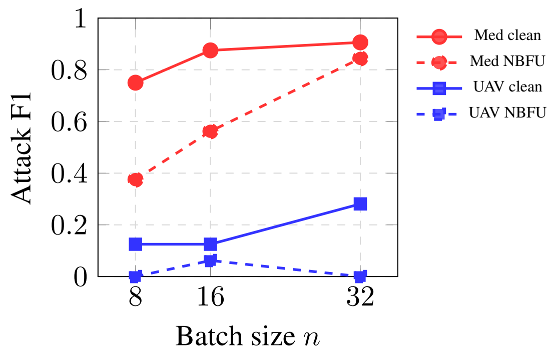

Figure 4: Attack F1 vs. batch size $n \in \{8, 16, 32\}$ (HF-GradInv, NBFU ε=5). Medical defense margin shrinks at larger n because the unclipped gradient norm decreases and the clean-baseline F1 ceiling rises. UAV remains fully defended across the range.

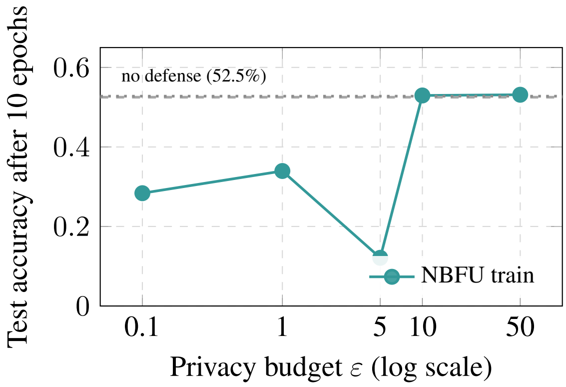

Figure 5: Privacy–utility trade-off under NBFU on ChestMNIST classification (companion to Table 3). Sharp cliff at $\varepsilon\approx$5–10. Dashed line: no-defense baseline. Pure clipping (σ=0, dotted) preserves utility.

D Language sweep — full table

E Frozen-backbone ablation

Our default setup keeps `requires_grad=True` on the ResNet-18 backbone, inflating d from the 148K-parameter head to the full 1.13×10^7. We rerun HF-GradInv on both domains with the backbone frozen so the client transmits only the 148K head gradient (3 seeds, sensitivity-$2C$ calibration). Medical clean F1 rises to 0.792±.07 (vs. 0.667 with the backbone, suggesting the head-only signal is somewhat *more* invertible); UAV clean F1 is unchanged (0.250±.13). NBFU at ε=5 drops medical to 0.333±.07 (58% reduction) and UAV to 0.042±.07 (83%); ε=10 gives the same numbers. The defense margin is therefore preserved or improved under the head-only transmission protocol. The SNR formula gives a higher predicted SNR at d=148K ($\sim$8.7×), but the empirical defense holds

Defense	Med. F1	UAV F1	Reliable
None (FedSGD)	0.667	0.250	—
Pruning 99%	inconsist.	inconsist.	no
Laplace noise	inconsist.	inconsist.	no
SimCLIP-4	0.542	0.250	no
FARE-4	0.375	0.292	no
FedAvg 5 epochs†	0.000	0.375	partial
FedAvg 20 epochs†	0.000	0.250	partial
FedAvg 5 + NBFU ε=5	0.500±.13	**0.000±.00**	combined
NBFU ε=5 (2C)	**0.292±.07**	**0.125±.22**	**yes**
NBFU ε=10 (2C)	**0.333±.12**	**0.042±.06**	**yes**

Table 7: Defense comparison against Geminio (HF-GradInv). 3-seed means for all rows except plain FedAvg (†, single-seed). Continuous $\overline{\cos}$ at NBFU ε=5 is 0.47 (medical) and 0.59 (UAV), vs. 0.92/0.88 clean.

because the noise is still added to all 148K head-gradient dimensions at the same σ and the gradient-matching objectives are largely scale-invariant; the σ=0 mechanism analysis remains the operative explanation.

F Suspect-round detection candidates

A client running NBFU conditionally needs a local detector. Three candidate signals are visible client-side without server cooperation: (i) *Parameter-divergence between consecutive received models*: an active server sending a malicious θ_Q produces a much larger $\|\theta_t - \theta_{t-1}\|$ than benign FedAvg, since θ_Q is specifically perturbed to amplify gradients on query-matching samples; (ii) *Per-sample loss skew on the client's auxiliary holdout*: θ_Q produces a long-tailed loss distribution where a small subset (the targeted samples) has anomalously high loss; (iii) *Output-layer rank or norm anomalies*: the rigged classifier head's weights have non-typical singular-value spectra. Any one signal can trigger conditional ε=5 NBFU. A full study—with false-positive analysis on benign FL rounds—is left to future work.

G End-to-end VLM table

Per-VLM end-to-end attack F1 (HF-GradInv, no defense), 3-seed mean±std, with paired t-test significance markers against vanilla CLIP (df=2): * $|t|>2.0$, ** $|t|>2.92$ (one-tail $p<0.05$).

Tag	Class	Loss ratio R	Discrim. Δ	Clean F1	NBFU ε=5 F1
Medical (BiomedCLIP)					
pna_base	paraphrase	9.45	0.084	0.917±.12	0.750±.10
pna_terse	paraphrase	5.48	0.081	0.583±.12	0.375±.00
pna_verbose	paraphrase	11.21	0.092	0.875±.00	0.458±.06
pna_synonym	paraphrase	3.17	0.083	0.125±.00	0.208±.12
pna_indirect	paraphrase	3.05	0.065	0.458±.06	0.583±.06
pna_compAND	composition	11.36	0.109	0.333±.06	0.708±.06
pna_compOR	composition	5.12	0.103	0.208±.12	0.292±.06
pna_negEX	negation	5.48	0.077	0.542±.12	0.542±.06
pna_negPURE	negation	5.52	0.097	0.333±.12	0.042±.06
pna_es	diversity	6.01	0.083	0.500±.00	0.292±.06
pna_typo	diversity	11.44	0.139	0.583±.06	0.708±.12
pna_xlong	diversity	10.67	0.088	0.750±.00	0.500±.18
UAV (CLIP)					
sol_base	paraphrase	130.54	0.078	0.292±.06	0.083±.06
sol_terse	paraphrase	165.36	0.070	0.417±.06	0.083±.06
sol_verbose	paraphrase	114.65	0.084	0.167±.12	0.000±.00
sol_synonym	paraphrase	52.84	0.080	0.417±.06	0.042±.06
sol_indirect	paraphrase	161.54	0.059	0.417±.06	0.042±.06
sol_compAND	composition	92.27	0.101	0.375±.10	0.042±.06
sol_compOR	composition	40.27	0.062	0.333±.12	0.083±.12
sol_negEX	negation	93.15	0.063	0.375±.00	0.125±.18
sol_negPURE	negation	44.81	0.056	0.500±.00	0.083±.06
sol_es	diversity	36.63	0.060	0.417±.06	0.000±.00
sol_typo	diversity	31.03	0.062	0.333±.06	0.125±.10
sol_xlong	diversity	106.26	0.090	0.333±.06	0.000±.00

Table 8: Language sweep across 24 queries (12 medical + 12 UAV). Loss ratio R is the post-training amplification factor; discriminability Δ is the top-10%–bottom-10% cosine-similarity margin of the VLM text embedding against the auxiliary image set. Clean F1 and NBFU ε=5 F1 are mean attack F1 (HF-GradInv, 24k iterations) over 3 seeds (42, 123, 256). On UAV, NBFU drives F1 to $\leq$0.13 across all 12 queries (mean 0.06); on medical, defense margins vary from $+88\%$ to -113% per query, indicating that defense robustness depends on query phrasing in the visually-similar domain.

VLM	Medical F1	UAV F1
Vanilla CLIP	**0.750±.13**	0.250±.13
SimCLIP-2	0.250±.00**	0.375±.13**
SimCLIP-4	0.542±.19	0.250±.00
FARE-2	0.625±.13**	0.417±.19
FARE-4	0.375±.13*	0.292±.19
TeCoA-2	0.333±.19**	0.250±.00
TeCoA-4	0.542±.07*	**0.458±.07****

Table 9: On medical, 3 of 6 robust variants weaken the attack at $p < 0.05$ (SimCLIP-2, FARE-2, TeCoA-2) and two more are marginal (*). On UAV only 2 of 6 reach significance (SimCLIP-2 weakens, TeCoA-4 strengthens); the remaining four robust variants fall within seed variance. With n=3 seeds the power is limited.